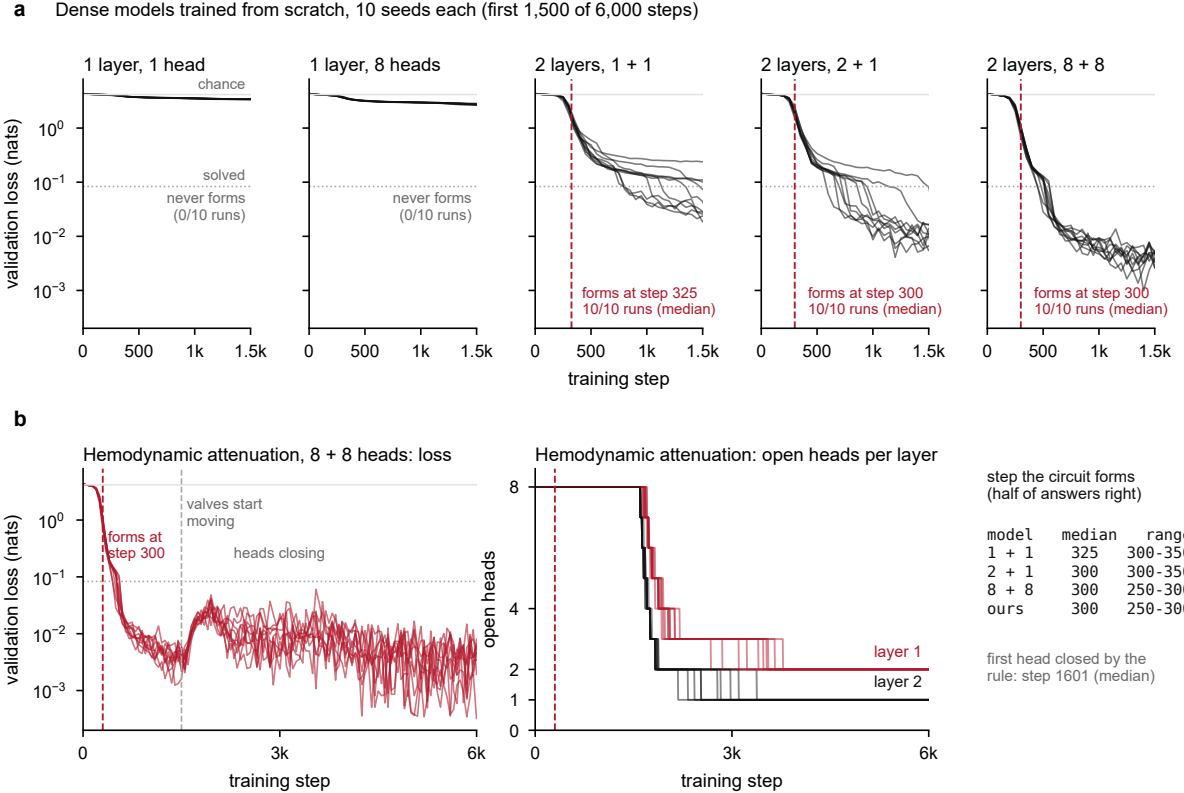

a
Dense models trained from scratch, 10 seeds each (first 1,500 of 6,000 steps)
1 layer, 1 head
1 layer, 8 heads
2 layers, 1 + 1
2 layers, 2 + 1
2 layers, 8 + 8
chance
solved
never forms
(0/10 runs)
never forms
(0/10 runs)
forms at step 325
10/10 runs (median)
forms at step 300
10/10 runs (median)
forms at step 300
10/10 runs (median)
validation loss (nats)
training step
b
Hemodynamic attenuation, 8 + 8 heads: loss
valves start
moving
forms at
step 300
heads closing
Hemodynamic attenuation: open heads per layer
open heads
layer 1
layer 2
step the circuit forms
(half of answers right)
model median range
1 + 1 325 300-350
2 + 1 300 300-350
8 + 8 300 250-300
ours 300 250-300
first head closed by the
rule: step 1601 (median)